

Deep Learning

Assignment 9

Implementation of RNN model for Stock Price Prediction

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```
pip install yfinance
```

```
Requirement already satisfied: yfinance in /usr/local/lib/python3.12/dist-packages (0.2.66)
Requirement already satisfied: pandas>=1.3.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.2.2)
Requirement already satisfied: numpy>=1.16.5 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.0.2)
Requirement already satisfied: requests>=2.31 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.32.4)
Requirement already satisfied: multitasking>=0.0.7 in /usr/local/lib/python3.12/dist-packages (from yfinance) (0.0.12)
Requirement already satisfied: platformdirs>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.9.6)
Requirement already satisfied: pytz>=2022.5 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2025.2)
Requirement already satisfied: frozendict>=2.3.4 in /usr/local/lib/python3.12/dist-packages (from yfinance) (2.4.7)
Requirement already satisfied: peewee>=3.16.2 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.0.4)
Requirement already satisfied: beautifulsoup4>=4.11.1 in /usr/local/lib/python3.12/dist-packages (from yfinance) (4.13.5)
Requirement already satisfied: curl_cffi>=0.7 in /usr/local/lib/python3.12/dist-packages (from yfinance) (0.15.0)
Requirement already satisfied: protobuf>=3.19.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (5.29.6)
Requirement already satisfied: websockets>=13.0 in /usr/local/lib/python3.12/dist-packages (from yfinance) (15.0.1)
Requirement already satisfied: soupsieve>1.2 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4>=4.11.1->yfinance) (2.7)
Requirement already satisfied: typing-extensions>=4.0.0 in /usr/local/lib/python3.12/dist-packages (from beautifulsoup4>=4.11.1->yfinance) (4.13.2)
Requirement already satisfied: cffi>=2.0.0 in /usr/local/lib/python3.12/dist-packages (from curl_cffi>=0.7->yfinance) (2.0.0)
Requirement already satisfied: certifi>=2024.2.2 in /usr/local/lib/python3.12/dist-packages (from curl_cffi>=0.7->yfinance) (2024.8.30)
Requirement already satisfied: rich in /usr/local/lib/python3.12/dist-packages (from curl_cffi>=0.7->yfinance) (13.9.4)
Requirement already satisfied: python-dateutil>=2.8.2 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->yfinance) (2.9.0)
Requirement already satisfied: tzdata>=2022.7 in /usr/local/lib/python3.12/dist-packages (from pandas>=1.3.0->yfinance) (2025.1)
Requirement already satisfied: charset-normalizer<4,>=2 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->yfinance) (3.4.0)
Requirement already satisfied: idna<4,>=2.5 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->yfinance) (3.11)
Requirement already satisfied: urllib3<3,>=1.21.1 in /usr/local/lib/python3.12/dist-packages (from requests>=2.31->yfinance) (2.3.0)
Requirement already satisfied: pycparser in /usr/local/lib/python3.12/dist-packages (from cffi>=2.0.0->curl_cffi>=0.7->yfinance) (2.0)
Requirement already satisfied: six>=1.5 in /usr/local/lib/python3.12/dist-packages (from python-dateutil>=2.8.2->pandas>=1.3.0->yfinance) (1.17.0)
Requirement already satisfied: markdown-it-py>=2.2.0 in /usr/local/lib/python3.12/dist-packages (from rich->curl_cffi>=0.7->yfinance) (3.0.0)
Requirement already satisfied: pygments<3.0.0,>=2.13.0 in /usr/local/lib/python3.12/dist-packages (from rich->curl_cffi>=0.7->yfinance) (2.18.1)
Requirement already satisfied: mdurl~=0.1 in /usr/local/lib/python3.12/dist-packages (from markdown-it-py>=2.2.0->rich->curl_cffi>=0.7->yfinance) (0.1.2)
```

```
import pandas as pd
import numpy as np
from sklearn.preprocessing import MinMaxScaler
from tensorflow.keras.models import Sequential
from tensorflow.keras.layers import Dense, LSTM
import yfinance as yf
```

```
data= yf.download("AAPL",start="2015-01-01" , end="2023-01-01")
df = data[['Close']]
df.head()
```

```
scaler=MinMaxScaler(feature_range=(0,1))
scaled_data=scaler.fit_transform(df)
```

```
X=[]
y=[]
```

```
/tmp/ipykernel_6200/1677913508.py:8: FutureWarning: YF.download() has changed argument auto_adjust default to True
data= yf.download("AAPL",start="2015-01-01" , end="2023-01-01")
[*****100%*****] 1 of 1 completed
```

```
import matplotlib.pyplot as plt
```

```
time_step = 60
```

```
for i in range(time_step,len(scaled_data)):
    X.append(scaled_data[i-time_step:i, 0])
    y.append(scaled_data[i, 0])
X, y = np.array(X), np.array(y)
X = X.reshape(X.shape[0], X.shape[1], 1)
train_size = int(len(X) * 0.8)
```

```
X_train = X[:train_size]
X_test = X[train_size:]
```

```

y_train = y[:train_size]
y_test = y[train_size:]

model = Sequential()
model.add(LSTM(units=50, return_sequences=True, input_shape=(X.shape[1],1)))
model.add(LSTM(units=50))
model.add(Dense(1))
model.compile(
    optimizer='adam',
    loss='mean_squared_error')
model.fit(
    X_train,
    y_train,
    epochs=10,
    batch_size=32)

```

```

/usr/local/lib/python3.12/dist-packages/keras/src/layers/rnn/rnn.py:199: UserWarning: Do not pass an `input_shape`/`input_dim` argument to the layer constructor, as they will be ignored. Instead, pass them to the `compile` method of the model.
  super().__init__(**kwargs)

```

```

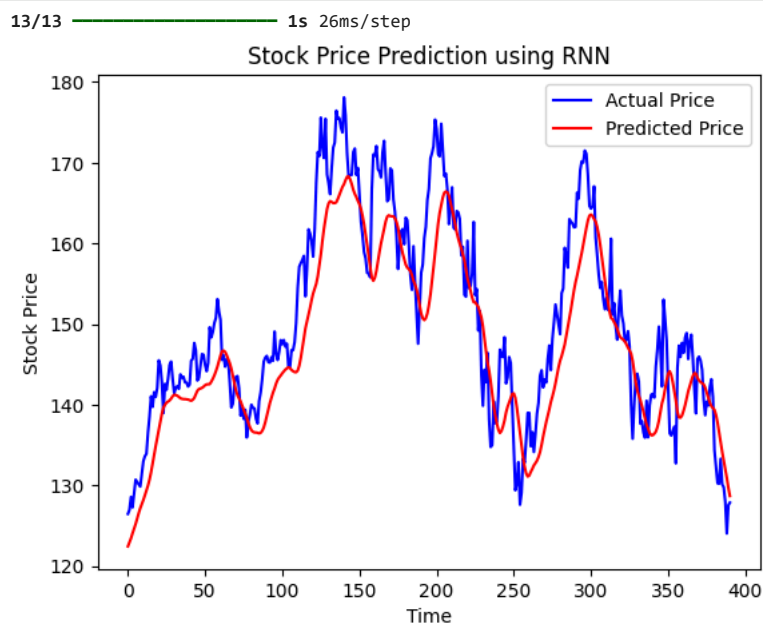
Epoch 1/10
49/49 ————— 5s 41ms/step - loss: 0.0089
Epoch 2/10
49/49 ————— 4s 72ms/step - loss: 3.2497e-04
Epoch 3/10
49/49 ————— 3s 59ms/step - loss: 2.8552e-04
Epoch 4/10
49/49 ————— 4s 31ms/step - loss: 2.8490e-04
Epoch 5/10
49/49 ————— 2s 32ms/step - loss: 2.6925e-04
Epoch 6/10
49/49 ————— 2s 31ms/step - loss: 2.7373e-04
Epoch 7/10
49/49 ————— 2s 31ms/step - loss: 2.7359e-04
Epoch 8/10
49/49 ————— 2s 34ms/step - loss: 2.6165e-04
Epoch 9/10
49/49 ————— 2s 38ms/step - loss: 2.4359e-04
Epoch 10/10
49/49 ————— 2s 31ms/step - loss: 2.2806e-04
<keras.src.callbacks.history.History at 0x78b774f7f080>

```

```

predictions = model.predict(X_test)
predictions = scaler.inverse_transform(predictions.reshape(-1,1))
y_test_actual = scaler.inverse_transform(y_test.reshape(-1,1))
plt.plot(y_test_actual, color='blue', label='Actual Price')
plt.plot(predictions, color='red', label='Predicted Price')
plt.title("Stock Price Prediction using RNN")
plt.xlabel("Time")
plt.ylabel("Stock Price")
plt.legend()
plt.show()

```



```

from sklearn.metrics import mean_squared_error
mse = mean_squared_error(y_test_actual, predictions)

```

```
rmse = np.sqrt(mse)
print("MSE:", mse)
print("RMSE:", rmse)
```

```
MSE: 39.294223208350516
RMSE: 6.268510445739922
```